

Validation of a digital, tablet-based version of the Trail Making Test in the Δ elta platform

Ebru Baykara¹  | Caroline Kuhn² | Nicklas Linz³ | Johannes Tröger^{3,4} | Julia Karbach¹

¹Department of Psychology, University of Koblenz-Landau, Landau, Germany

²Clinical Neuropsychology and Neuropsychological Outpatient Unit, Saarland University, Saarbrücken, Germany

³ki elements UG, Saarbrücken, Germany

⁴German Research Center for Artificial Intelligence GmbH, Kaiserslautern, Germany

Correspondence

Ebru Baykara, University of Koblenz-Landau, Fortstr. 7, 76829 Landau, Germany.
Email: baykara@uni-landau.de

Funding information

University of Koblenz-Landau

Edited by: Ali Mazaheri

Abstract

Using digital technology for neuropsychological assessment is gaining popularity in both clinical and research settings. Digital neuropsychology offers many benefits over the traditional paper-pencil assessments; however, their comparability requires further validation. The aim of this study was to compare a digital, tablet-based Trail Making Test to the standard paper version. In a within-subject design, 108 healthy adults completed both digital and paper Trail Making Test in a counterbalanced order. Each participant also performed other tasks measuring core executive abilities (inhibition, working memory, and flexibility) on the tablet. Our findings indicated that the Trail Making Test performance on the two different modalities correlated significantly. Furthermore, correlations of Trail Making Test performance with other cognitive tasks revealed that digital Trail Making Test is comparable with the paper version. However, the modality had a significant effect on Trail Making Test performance; that is, participants were generally faster on the digital platform. Taken together, our findings suggest that with new normative data, traditional Trail Making Test can be adapted successfully to a digital platform.

KEYWORDS

computerized assessment, digital neuropsychology, digitization of neuropsychological assessment, mobile application, paper- and tablet-based comparison, Trail Making Test

1 | INTRODUCTION

With the rise and advancement of digital technology, digital neuropsychology is becoming more popular in both research and clinical settings. Using technology for neuropsychological assessment is attractive for many reasons: It can provide new measures that are not available in traditional tests, it can enable testing outside of clinical/research settings thereby increasing ecological validity, it can allow more standardized and precise

measurements. Digitization can also be cost-effective: Neuropsychological assessments require trained personnel and are very costly and time consuming. Digital assessments can require less supervision and allow more individuals to be tested and more frequently.

Although they have many advantages, digital methods are not necessarily comparable with their traditional counterparts, and certain precautions are needed when digitizing traditional paper-pencil assessments. Traditional methods are established by years of research, validation, standardization, and fine-tuning, whereas digital methods still need to be psychometrically evaluated and validated.

Abbreviation: TMT, Trail Making Test.

Since the benefits of using technology in neuropsychology exceed associated problems and challenges, there is a growing number of studies that aim to establish the validity and reliability of digital assessment methods (for an elaborated discussion of challenges and opportunities related to digital neuropsychology, see Germine et al., 2019).

One of the most widely used neuropsychological tests is the Trail Making Test (TMT, Reitan & Wolfson, 1985). This is due to the simplicity of its administration and its high sensitivity to cognitive dysfunction. TMT can measure a variety of cognitive processes including processing speed, visuo-motor skills, and executive function, such as working memory and cognitive flexibility (Sánchez-Cubillo et al., 2009). Due to its extensive use in both clinical and research settings and its inclusion in most neuropsychological test batteries, it is important to evaluate and establish the validity of a digitized version. TMT results are evaluated according to age and education stratified norms, and it is important to assess how a digital TMT compares to the paper-pencil version.

A digitized version of TMT has been compared with the traditional paper-pencil version previously; however, the results were inconclusive. Whereas some studies indicated that digital and paper versions are adequately comparable (Björngrim et al., 2019; Dahmen et al., 2017; Fellows et al., 2017), others reported significant differences in the measures from different modalities (Bracken et al., 2018; Rodriguez et al., 2019). The aim of the current study was to compare participant's performance on digital and paper-pencil versions of TMT and additionally to evaluate the validity of the measures by (1) comparing the correlations between performance on the digital and the paper-pencil version of the TMT, (2) comparing performance on both TMT versions to performance on tests measuring related executive control abilities (working memory, inhibition, and cognitive flexibility; cf. Miyake et al., 2000), and (3) comparing the size of the correlations between both measures of the TMT and the related executive function tasks. We expected substantial correlations between performances on both TMT tasks and medium-sized correlations with performance on other executive control tasks. The size of these correlations was not expected to differ between the digital and the paper-pencil version of the TMT. Furthermore, we investigated the effects of age, gender, and education on both TMT measures.

2 | METHODS

2.1 | Study design

A total of 108 healthy young adults (age: $M = 24.49$ years, $SD = 6.82$; education in years: $M = 15.48$,

$SD = 2.38$; gender: 78.7% female) from the universities of Koblenz-Landau and Saarland participated in this study (89% right-handed). All participants completed a battery of neuropsychological tests, all of which (except the paper-based TMT test) were administered via a tablet (iPad Pro 12.9", 3. Gen, iOS Version 11.4.1) using the Δ elta¹ cognitive testing platform (Version 1.1) by a trained research assistant in a lab environment at the university. Delta was an iPad app designed to facilitate neurocognitive testing by digitalizing and partly automating common paper pencil neurocognitive assessments, such as the TMT. It included a test battery with all the neurocognitive assessments included in this study. The tests were started by an interface allowing to enter the participant identification number and basic demographic data, followed by a standardized presentation of the neurocognitive tests.

Prior to participation, all subjects received written information about the study and storage and usage of data. All participants signed a written informed consent form. The study was conducted in accordance with standards of the university ethics committees and the Helsinki Declaration.

3 | MATERIALS AND PROCEDURE

First, participants entered their sociodemographic data (gender, age, years of education, and handedness) in the tablet. Then, half of the participants randomly started with the paper-based and the other half with the digital version of TMT tests at the beginning of the session. The order of the following neuropsychological tests (working memory, inhibition, flexibility) was counterbalanced between participants. During the study, participants were sitting at a table with the tablet lying flat on the surface, either vertically or horizontally positioned, depending on the test at hand. For each test, they were instructed by the experimenter and they also read the instructions on the tablet screen.

3.1 | Executive functions: Trail Making Test (cf. Reitan & Wolfson, 1985)

Both paper and digital versions of the TMT tests consisted of 25 circles. In TMT-A, the task was to connect the 25 numbers in a numerical order. In TMT-B, which comprises both numbers and letters, the task was to connect items alternately in an ascending (numerical and alphabetical) order. The size, position, and distance of

¹<https://ki-elements.de/en/diagnosis-support/>

the items were identical in the paper and digital versions. Participants had one practice trial (eight items) before each version of TMT-A and B. For the paper version of the TMT, the experimenter recorded the time to complete the tests, starting when the participant touched the first item on the sheet and stopping when the last item was reached. In case of errors, the experimenter notified the participants and let them correct their errors. In the tablet version of the tests, participants used a stylus (Apple Pencil, 1. Gen). Measurement of the response time also started as the participants touched the first item on the screen and stopped when the last item was reached. If the participant made an error, the incorrect line would turn red, and participants needed to correct their mistake as in the paper-based version. For both versions of the tests, the dependent measure was the time to complete the task. The number of errors was not analysed separately given that it is reflected in the completion time (because participants are stopped and returned to the last correct response if they commit an error; Bowie & Harvey, 2006).

3.2 | Working memory: Corsi Block Tapping Backward test (cf. Corsi, 1972)

There were in total nine blue blocks at a fixed spatial arrangement on the tablet screen. The blocks were lit in sequences of increasing length (from two to nine blocks), and two trials with different sequences (of equal length) were performed at each level. Stimuli were lit for 1000 ms. The same block was not lit twice within a trial. The participant's task was to repeat the order backwards. If participants made any errors, the number of blocks lit remained the same. If two trials with the same number of blocks were successfully completed, that item number was increased by 1; otherwise, the task was stopped. Participants had three practice trials (two-, three-, and four-block sequences) prior to the task. Neither practice nor test trials were provided with feedback. The dependent measure in this task was the longest sequence the participant could correctly repeat.

3.3 | Inhibition: Go/No-Go test (cf. Wessel, 2018)

Participants saw pictures on the screen and were required to tap the screen when the stimulus was a flower, and to not react when it was a tree. The stimuli were selected randomly from four different flower and four different tree pictures. There were in total 30 trials, of which 80%

included go stimuli and 20% no-go stimuli. Participants had 10 practice trials before the actual test. The stimulus duration was set at 1500 ms, and interstimulus interval was 50 ms. No feedback was provided. The dependent variables were the reaction time in the correct Go trials and the error rate in the No-Go trials.

3.4 | Flexibility: Word fluency task (cf. Aschenbrenner et al., 2000)

In the switching version of the semantic fluency test, participants named as many items as possible from the categories of sports and fruits in 60 s. The words said twice or stemming from the same root were counted only once. Number of switches was used as the dependent measure. There was no practice, but the experimenter gave examples from other categories to clarify the instructions.

3.5 | Statistical analysis

Statistical analyses were performed using SPSS Statistics (version 26.0). Comparisons between the paper and digital versions of the TMT test were carried out using correlations (Spearman's rho for non-normally distributed data). We used repeated measures ANOVA to analyse the effects of order of tests and test modality (paper vs. digital). To test for the effects of age, gender, and years of education on the TMT scores, we employed regression analyses. In order to analyse the association between both the paper version and digital version of the TMT tests and the working memory, inhibition, and flexibility tests, we employed Spearman rank correlation analyses. We compared the strength of overlapping correlations based on the method by Hittner et al. (2003) and nonoverlapping correlations based on Silver et al. (2004).

4 | RESULTS

4.1 | Correlation between digital and paper TMT performances

The completion times of both TMT-A (paper: $M = 23.05$ s, $SD = 6.12$, digital: $M = 21.08$ s, $SD = 5.60$; $r_s = 0.379$, $p < 0.001$) and TMT-B (paper: $M = 46.75$ s, $SD = 16.47$, digital: $M = 41.73$ s, $SD = 12.66$; $r_s = 0.431$, $p < 0.001$) correlated significantly between the paper and tablet versions (see Figure 1). The strength of the

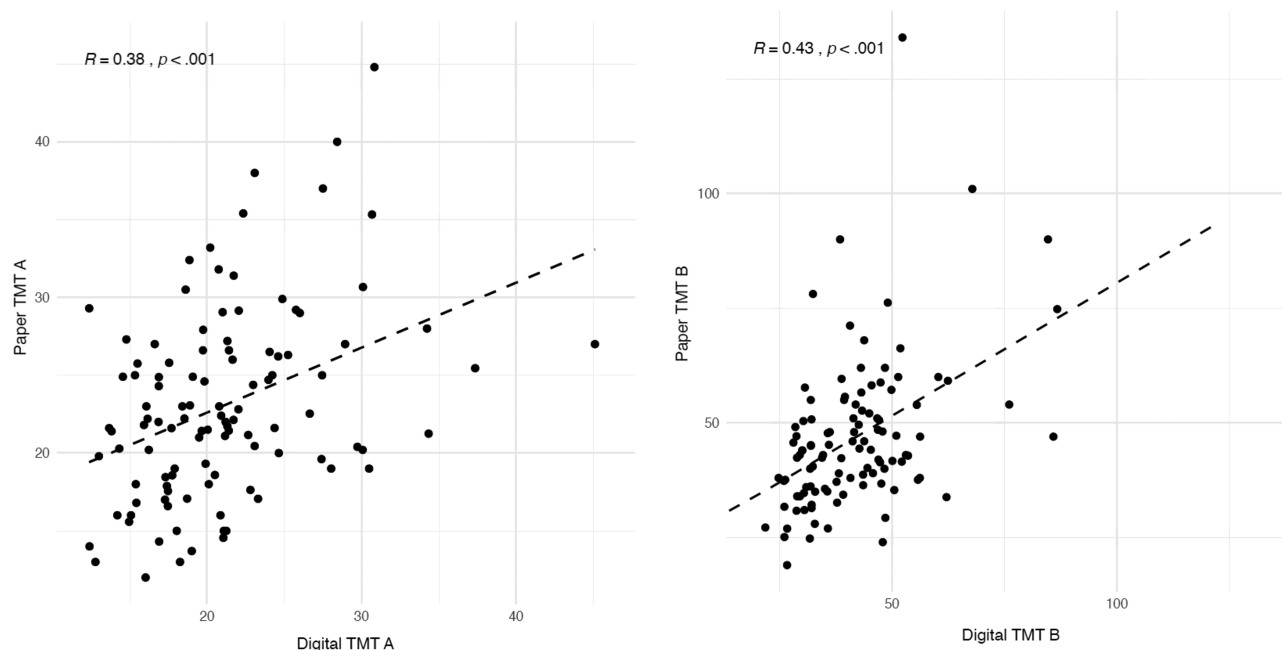


FIGURE 1 Correlations between digital and paper Trail-Making Tests. All values are in seconds

correlations did not significantly differ ($z = -0.53$, $p = 0.600$).

4.2 | Effect of assessment order and assessment modality on TMT performance

The assessment order (digital or paper first) had no significant effect on the completion times of TMT-A ($p = 0.452$) and TMT-B ($p = 0.976$). However, there were significant effects of assessment modality on both TMT-A, $F(1, 106) = 3.937$, $p = 0.050$, $\eta_p^2 = 0.036$, and TMT-B, $F(1, 106) = 5.127$, $p = 0.026$, $\eta_p^2 = 0.046$, completion times, indicating that participants completed the digital versions of the tests significantly faster. More importantly, there were significant interaction effects of order $\times$ modality on the completion times of both TMT-A, $F(1, 106) = 30.823$, $p < 0.001$, $\eta_p^2 = 0.225$, and TMT-B, $F(1, 106) = 24.675$, $p < 0.001$, $\eta_p^2 = 0.189$. Participants were significantly slower on the modality they started with and were faster in their second test (retest effect; see Figure 2).

4.3 | Effects of age, education, and gender on TMT performance

The results of multiple regression analyses revealed that there were no significant effects of age, education and gender on digital and paper TMT measures (p values ranging between 0.180 and 0.976).

4.4 | Correlations with other cognitive tasks

In order to test the validity of the TMT versions, we computed correlations between performance on the TMT and the Go/No-Go task (inhibition), the Corsi Span task (working memory), and the fluency task (flexibility). We will present the results in this task order and report correlations with digital TMT-A, paper TMT-A, digital TMT-B, and paper TMT-B, respectively. For all significant correlations, we also report whether they differ in size between the digital and the paper version.

4.4.1 | Go/No-Go task

Since the variation in the number of errors (%) in the Go/No-Go task was very small ($M = 0.034$, $SD = 0.085$; 80% of participants committed no errors), we excluded the errors from our analyses. Data for this task were missing for eight participants for technical reasons. The average reaction times to Go stimuli ($N = 100$, $M = 932.51$ ms, $SD = 89.25$) was significantly correlated with paper TMT-A ($r_s = 0.393$, $p < 0.001$), digital TMT-A ($r_s = 0.252$, $p = 0.012$), and paper TMT-B ($r_s = 0.271$, $p = 0.006$), but not with digital TMT-B ($r_s = 0.173$, $p = 0.085$). The correlation coefficients between Go RT and paper TMT-A and Go RT and digital TMT-A were not significantly different in strength ($z = 1.36$, $p = 0.174$).

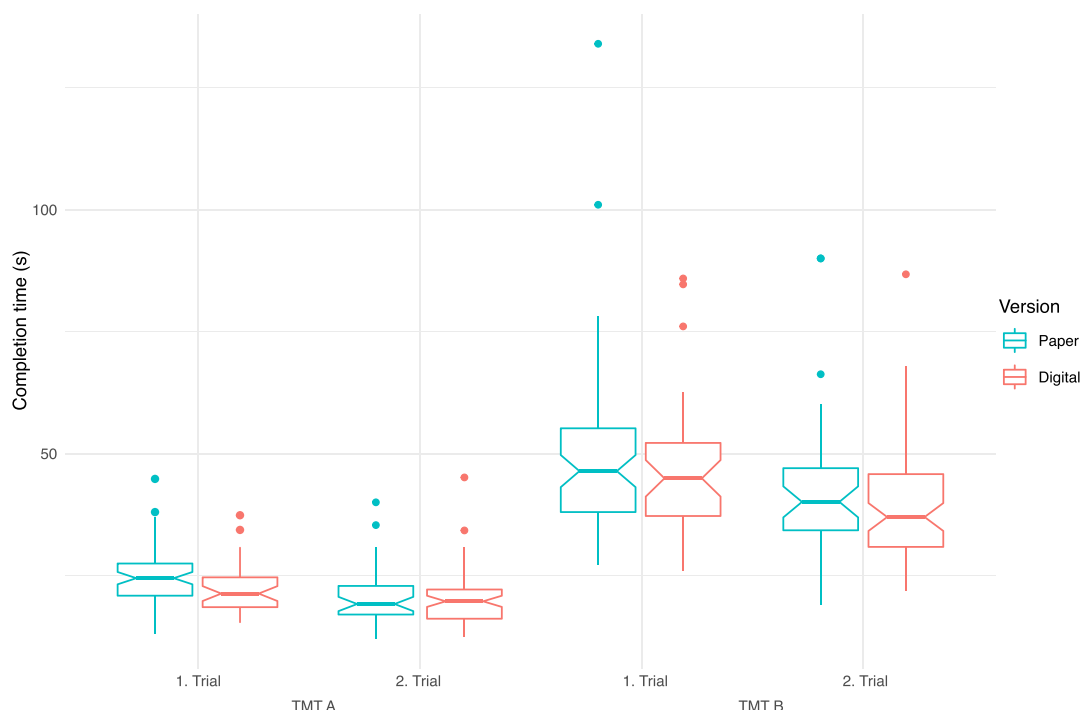


FIGURE 2 Performance on digital and paper Trail-Making Tests based on assessment order. Participants are faster at the second trials of tests independent of the modality. They are also on average faster in the digital modality

4.4.2 | Corsi Span task

The Corsi Span performance ($N = 108$, $M = 6.06$, $SD = 1.49$) correlated significantly with digital TMT-A ($r_s = -0.319$, $p < 0.001$) and marginally with paper TMT-A ($r_s = -0.178$, $p = 0.065$). The correlation coefficients did not significantly differ from each other ($z = -1.36$, $p = 0.175$). The correlation between Corsi Span and TMT-B performances was significant for both modalities (paper: $r_s = -0.237$, $p = 0.014$; digital: $r_s = -0.367$, $p < 0.001$), and the strength of the correlations for paper and digital TMT-B was not significantly different ($z = 1.33$, $p = 0.183$).

4.4.3 | Fluency task

Performance on the fluency task (number of switches; $N = 108$, $M = 15.60$, $SD = 2.90$) correlated significantly with the digital TMT-B ($r_s = -0.319$, $p < 0.001$), but not with paper TMT-B or any of the TMT-A versions (all p values > 0.1).

5 | DISCUSSION

The main focus of this study was to compare a digitized Trail Making Test (TMT) to the traditional paper-pencil

version. Our results are in line with the findings from earlier studies (e.g., Björngrim et al., 2019; Fellows et al., 2017) that showed a medium-sized correlation between the digitized and traditional versions of TMT. Our findings also revealed that a measure of processing speed (Go reaction time) correlated significantly with TMT-A in both modalities, and a measure of executive function (Corsi Span) correlated with TMT-B in both modalities. Importantly, the strength of these correlations did not differ significantly between the two different versions (digital vs. paper-pencil). However, another executive function measure (switching subtest of Semantic Fluency) only correlated with digital TMT-B. Taken together, it can be concluded that digitized TMT measures are comparable to the traditional TMT; nevertheless, the tests in different modalities are not parallel to each other and cannot be used interchangeably.

Digital adaptations of traditional tests can complete the paper-pencil assessments or even replace them in the future; however, the norms collected with traditional measures may not be applicable for digital assessments. In line with previous findings (e.g., Rodriguez et al., 2019), our results revealed that the modality of the assessment had a significant effect on completion times; that is, participants completed the digital version of the test significantly faster than the paper version. Studies with older adults (Latendorf et al., 2021; Steinert et al., 2018) found the opposite effect, which was

influenced by participants' familiarity with technology use. The participants in our study were mostly young university students, who are familiar with digital technology and therefore, a lack of familiarity with the tablet is unlikely to have influenced the results. Although not tested directly, it is plausible that the pressure needed to draw a line on the tablet with a stylus is lower than the pressure needed on paper with a pencil, which in turn can have an effect on how fast one can draw on different modalities. It is also possible that the digital recording on the tablet is more accurate than the completion times recorded by examiners, and consequently, the digital results are faster than the paper-pencil ones.

The findings of the current study suggest that digitized TMT is a valid adaptation of the traditional method; however, the norms for paper-pencil TMT are not applicable for the digital version. Standardized values must be collected specifically for the digitized assessments. It is costly and time consuming to collect normative data for the digital neuropsychological assessments; nevertheless, the benefits of using digital technology surpass the challenges (for a detailed evaluation, see Germine et al., 2019). Among other benefits, digital assessments can provide more accurate and detailed data and enable multiple measurements without the need of a trained personnel. An important problem for TMT is test-retest effect, which was also observed in this study: participants complete the test faster when they are tested again with the same test. With the help of digital technology, it could be possible to create new versions of the test while keeping the test properties (e.g., the distance between items) and test difficulty comparable between instances (Zeng et al., 2017) to minimize test-retest effects. Furthermore, digital assessment methods can be combined with neuroimaging (e.g., Karimpoor et al., 2017) and reduce test administrator influence on the assessment results (Overton et al., 2016). However, when interpreting the results of digital assessments, the variation between digital devices (software and hardware) and their possible impact on the measures are necessary to consider and to report.

Performance on TMT is typically influenced by age and education, and therefore, the normative data are stratified accordingly. In our study, we did not find any effect of age, education, or gender on TMT performance. It is a limitation of the current study that the participant sample mainly consisted of healthy young adults with relatively high educational levels, which is not a representative of the general population. However, given that this study aimed at validating the digital version of the TMT, we chose a homogenous group of participants. The next step will be to validate our findings in samples with various demographics, such as different age groups or

socioeconomic levels, as well as in samples with cognitive impairments.

6 | CONCLUSION

The field of neuropsychology should consider to increasingly incorporate and utilize the advances in digital technology. One way is to create assessment methods specifically designed for digital platforms. Although this has clear advantages, it is also very costly and time consuming. Another option is to adapt already existing and well-established methods to digital platforms, which requires validation and evaluation. The current study aimed at validating the feasibility of digitizing the Trail Making Test and our findings suggest that with new normative data, traditional TMT can be adapted successfully to a digital platform.

ACKNOWLEDGEMENT

This study was supported by a start-up grant from the University of Koblenz-Landau.

CONFLICT OF INTEREST

EB, CK, and JK declare no conflict of interest. NL and JT own stock in the digital biomarker company ki:elements.

AUTHOR CONTRIBUTIONS

JK, CK, NL, and JT designed the study. EB analysed the data and wrote the first draft of the manuscript. JK, CK, NL, and JT commented on this draft, and EB revised it accordingly.

PEER REVIEW

The peer review history for this article is available at <https://publons.com/publon/10.1111/ejn.15541>.

DATA AVAILABILITY STATEMENT

Data have been uploaded to OSF and can be accessed via the following link: https://osf.io/zrq26/?view_only=00069eda2b4d4b8492b6536adffc3a3a.

After publication of the paper, the data will be made publicly available on OSF.

ORCID

Ebru Baykara  <https://orcid.org/0000-0002-5841-8499>

REFERENCES

- Aschenbrenner, S., Tucha, O., & Lange, K. W. (2000). *RWT. Regensburger Wort-flussigkeits-Test*. Hogrefe-Verlag.
- Björngrim, S., van den Hurk, W., Betancort, M., Machado, A., & Lindau, M. (2019). Comparing traditional and digitized cognitive tests used in standard clinical evaluation—A study of the

- digital application minnemera. *Frontiers in Psychology*, 10, 2327. <https://doi.org/10.3389/fpsyg.2019.02327>
- Bowie, C. R., & Harvey, P. D. (2006). Administration and interpretation of the Trail Making Test. *Nature Protocols*, 1(5), 2277–2281. <https://doi.org/10.1038/nprot.2006.390>
- Bracken, M. R., Mazur-Mosiewicz, A., & Glazek, K. (2018). Trail making test: Comparison of paper-and-pencil and electronic versions. *Applied Neuropsychology: Adult*, 26(6), 522–532. <https://doi.org/10.1080/23279095.2018.1460371>
- Corsi, P. M. (1972). Human memory and the medial temporal region of the brain. *Dissertation Abstracts International*, 34, 819B.
- Dahmen, J., Cook, D., Fellows, R., & Schmitter-Edgecombe, M. (2017). An analysis of a digital variant of the trail making test using machine learning techniques. *Technology and Health Care*, 25(2), 251–264. <https://doi.org/10.3233/THC-161274>
- Fellows, R. P., Dahmen, J., Cook, D., & Schmitter-Edgecombe, M. (2017). Multicomponent analysis of a digital Trail Making Test. *The Clinical Neuropsychologist*, 31(1), 154–167. <https://doi.org/10.1080/13854046.2016.1238510>
- Germine, L., Reinecke, K., & Chaytor, N. S. (2019). Digital neuropsychology: Challenges and opportunities at the intersection of science and software. *The Clinical Neuropsychologist*, 33(2), 271–286. <https://doi.org/10.1080/13854046.2018.1535662>
- Hittner, J. B., May, K., & Silver, N. C. (2003). A Monte Carlo evaluation of tests for comparing dependent correlations. *The Journal of General Psychology*, 130, 149–168. <https://doi.org/10.1080/00221300309601282>
- Karimpoor, M., Churchill, N. W., Tam, F., Fischer, C. E., Schweizer, T. A., & Graham, S. J. (2017). Tablet-based functional MRI of the trail making test: Effect of tablet interaction mode. *Frontiers in Human Neuroscience*, 11, 496. <https://doi.org/10.3389/fnhum.2017.00496>
- Latendorf, A., Runde, L. M., Salminen, T., & Steinert, A. (2021). Digitization of neuropsychological diagnostics: A pilot study to compare three paper-based and digitized cognitive assessments. *Aging Clinical and Experimental Research*, 33, 1585–1597.
- Miyake, A., Friedman, N. P., Emerson, M. J., Witzki, A. H., Howerter, A., & Wager, T. D. (2000). The unity and diversity of executive functions and their contributions to complex “frontal lobe” tasks: A latent variable analysis. *Cognitive Psychology*, 41(1), 49–100. <https://doi.org/10.1006/cogp.1999.0734>
- Overton, M., Pihlgård, M., & Elmståhl, S. (2016). Test administrator effects on cognitive performance in a longitudinal study of ageing. *Cogent Psychology*, 3(1), 1260237. <https://doi.org/10.1080/23311908.2016.1260237>
- Reitan, R. M., & Wolfson, D. (1985). *The Halstead-Reitan neuropsychological test battery: Therapy and clinical interpretation*. Neuropsychological Press.
- Rodriguez, F. S., Spilski, J., Schneider, A., Hekele, F., Lachmann, T., Ebert, A., & Rupprecht, F. A. (2019). Relevance of the assessment mode in the digital assessment of processing speed. *Journal of Clinical and Experimental Neuropsychology*, 41(7), 730–739. <https://doi.org/10.1080/13803395.2019.1616079>
- Sánchez-Cubillo, I., Periañez, J. A., Adrover-Roig, D., Rodríguez-Sánchez, J. M., Rios-Lago, M., Tirapu, J. E. E. A., & Barcelo, F. (2009). Construct validity of the Trail Making Test: Role of task-switching, working memory, inhibition/interference control, and visuomotor abilities. *Journal of the International Neuropsychological Society: JINS*, 15(3), 438–450. <https://doi.org/10.1017/S1355617709090626>
- Silver, N. C., Hittner, J. B., & May, K. (2004). Testing dependent correlations with nonoverlapping variables: A Monte Carlo simulation. *Journal of Experimental Education*, 73, 53–69. <https://doi.org/10.3200/JEXE.71.1.53-70>
- Steinert, A., Latendorf, A., Salminen, T., & Müller-Werdan, U. (2018). Evaluation of technology-based neuropsychological assessments in older adults. *Innovation in Aging*, 2(1), 504. <https://doi.org/10.1093/geroni/igy023.1874>
- Wessel, J. R. (2018). Prepotent motor activity and inhibitory control demands in different variants of the go/no-go paradigm. *Psychophysiology*, 55(3), e12871. <https://doi.org/10.1111/psyp.12871>
- Zeng, Z., Miao, C., Leung, C., & Shen, Z. (2017). Computerizing trail making test for long-term cognitive self-assessment. *International Journal of Crowd Science*, 1(1), 83–99. <https://doi.org/10.1108/IJCS-12-2016-0002>

How to cite this article: Baykara, E., Kuhn, C., Linz, N., Tröger, J., & Karbach, J. (2022). Validation of a digital, tablet-based version of the Trail Making Test in the Delta platform. *European Journal of Neuroscience*, 55(2), 461–467. <https://doi.org/10.1111/ejn.15541>